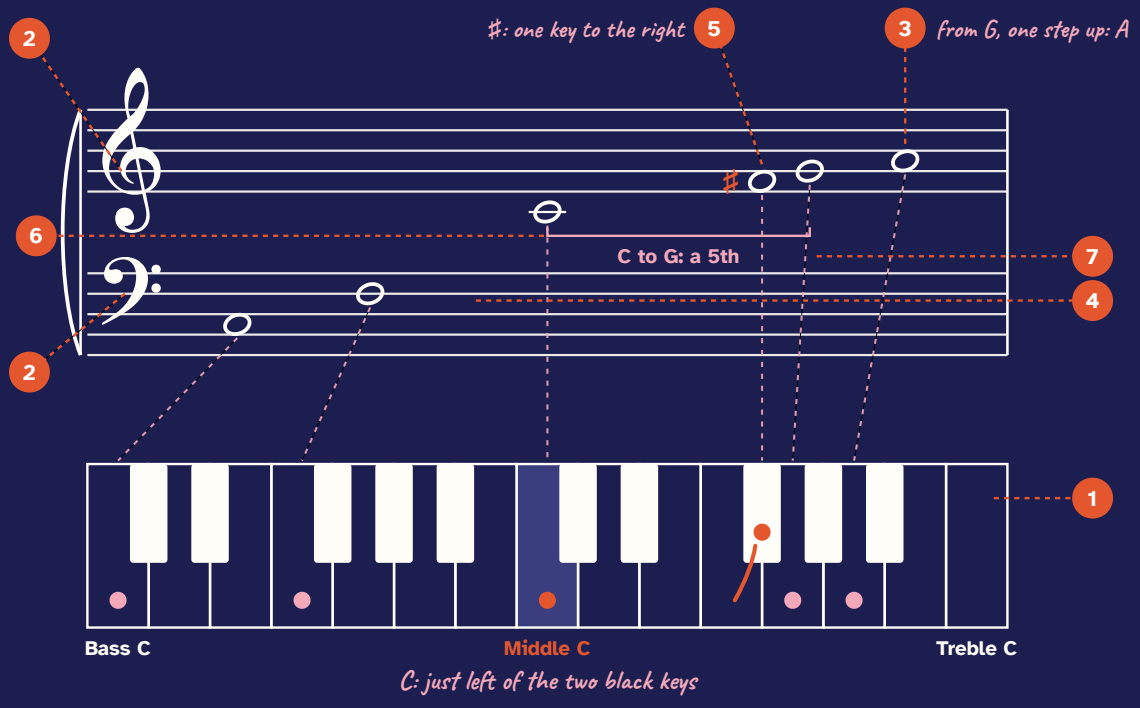
PART I · WHICH NOTE

Pitch

High or low? That is the first question every note answers. Start with the keys under your fingers, find where they live on the staff, see how sharps and flats move them, then measure how far apart two notes are. Everything in this part is in this one picture.



1 **The keyboard** p. 22
two and three black keys; C is left of the two

2 **The staff and the clefs** p. 25
the treble clef names G, the bass clef F

3 **Reading the notes** p. 28
start from a clef note, then step or skip

4 **The staff and the keyboard** p. 31
every line and space is one white key

5 **Sharps, flats and naturals** p. 34
a sharp moves one key right, a flat one key left

6 **Counting intervals** p. 37
count the notes, both ends: C to G is a 5th

7 **Major, minor and perfect** p. 40
count the hops: 7 make a perfect 5th

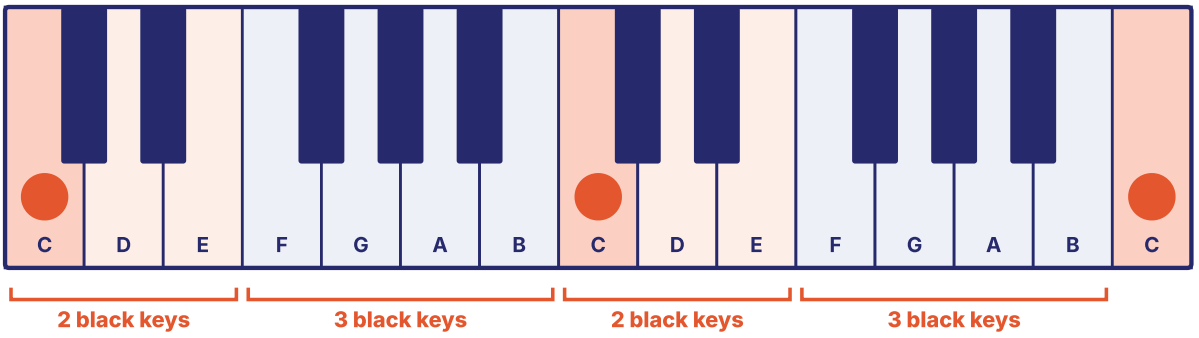
★ **Tips: when a note reads wrong** p. 43

PART I · PITCH · 1

The keyboard

Left is **low**, right is **high**. The black keys come in groups of **two** and **three**, again and again: that pattern is your map to every key.

THE SECRET
Find the two black keys: C is always just to their left.



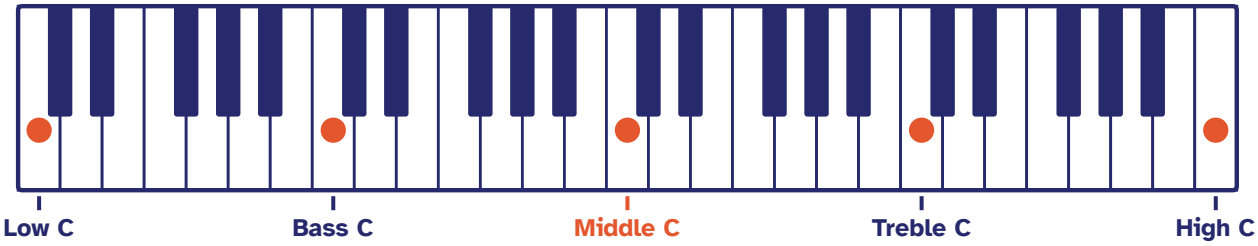
Every C sits just left of the two black keys

Seven letters, round and round

The white keys use seven letters, **A B C D E F G**, then start again, like the days of the week. **C D E** sit round the two black keys (pink), **F G A B** round the three (blue).

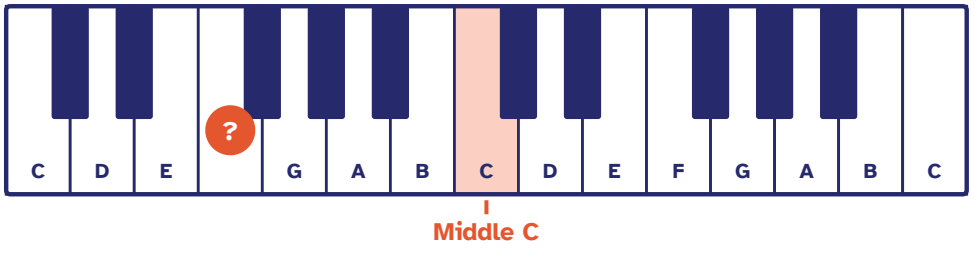
The five Cs have names

Find the C nearest the middle: that is **Middle C**. The Cs either side are **Bass C** and **Treble C**; further out, **Low C** and **High C**. From one C to the next is an **octave**: count the notes, C D E F G A B C, eight, both Cs included.



EXAMPLE 2

Find the F below Middle C



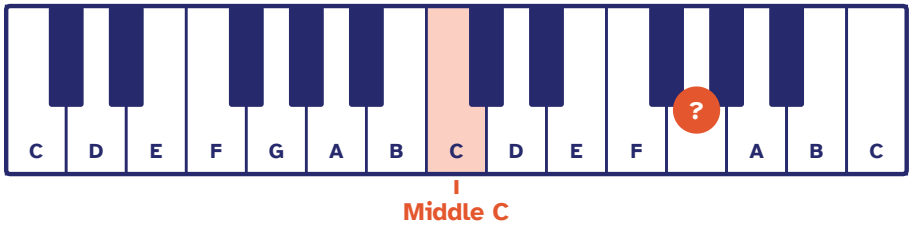
1. **Start on Middle C.**
2. **Look left** for the nearest group of **three** black keys.
3. **The white key just left of that group** is the **?**.

The **?** is **F**. Every F sits just left of three black keys, the way every C sits just left of two.

PART I · PITCH · 1 · CONTINUED

EXAMPLE 1

Find the G above Middle C

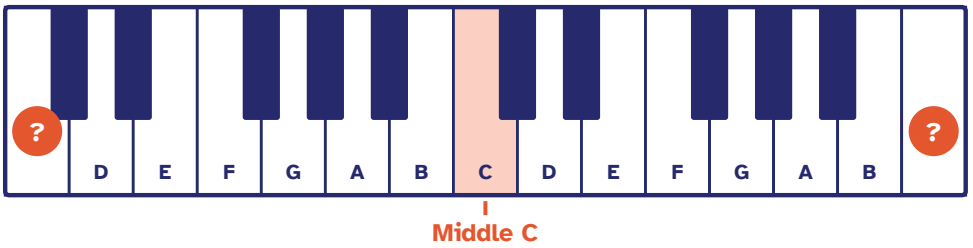


1. **Find Middle C:** the C just left of the two black keys nearest the middle.
2. **Walk up the letters**, one white key each, and count the notes: C is 1, then D 2, E 3, F 4 …
3. **Stop on 5:** that key is the **?**. Look where it sits.

The **?** is **G**, note 5: tucked inside the group of three black keys, between the first two.

EXAMPLE 3

Find Bass C and Treble C

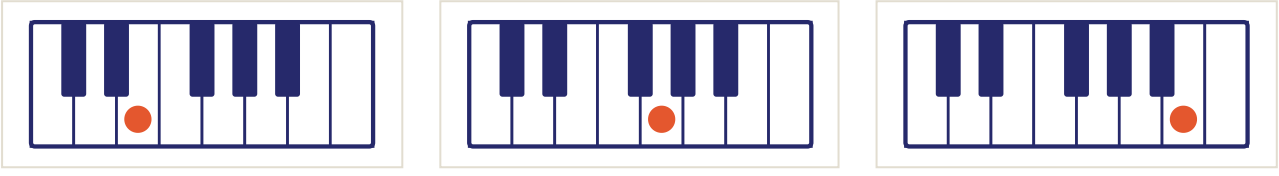


1. **Start on Middle C.**
2. **Walk left** to the next pair of black keys. The C just left of it is one **?**: count the notes, C B A G F E D C, eight.
3. **Walk right** the same way, to the next pair: the other **?**.

Left **?**: **Bass C**. Right **?**: **Treble C**. Each is one octave from Middle C.

GAME

Name the key



1 2 3

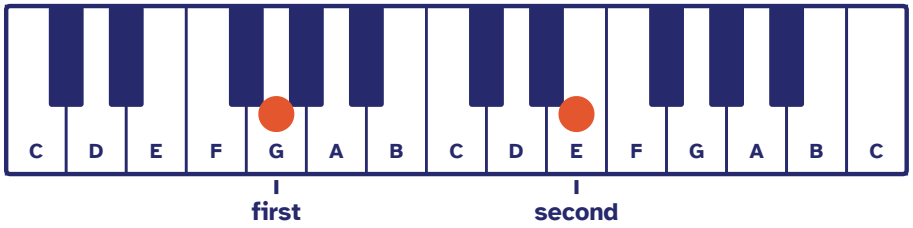
Each keyboard starts on C. Use the groups of black keys to name the red key.

Answer: 1 E (just right of the two) · 2 G (inside the three, after F) · 3 B (just right of the three).

PART I · PITCH · 1 · CONTINUED

EXAMPLE 4

Higher or lower?



1. **Find the two red keys.**
2. **Which one is further right?**
3. **Right means higher:** play both and listen.

The second note is **higher**: it is to the right. Left is always lower, right always higher.

TRY IT · 10 SECONDS

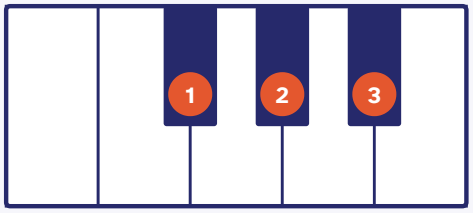


Play every C, bottom to top.

Say its name as you play it: Low C, Bass C, Middle C, Treble C, High C. Then play every F: left of the three.

YOU KNOW THIS

Hot Cross Buns, on three black keys



Play the group of three black keys from the top: **3 2 1**, 3 2 1, then 1 1 1 1, 2 2 2 2, 3 2 1. You just played a whole tune without one white key.

Try it. Find another group of three, higher up, and play it again there.

THINK OF IT The black keys are bumps you can feel in the dark. Close your eyes, slide your hand along, feel two black keys: the white key just to their left is C. Three black keys: the key to their left is F.

TRAP

× Mixing up C and F: C is left of the **two** black keys, F left of the **three**.

SEE ALSO

The staff and the clefs p. 25 · The staff and the keyboard p. 31

PRACTISE IT

Workbook Pitch 1 · Online: Library › Keyboard

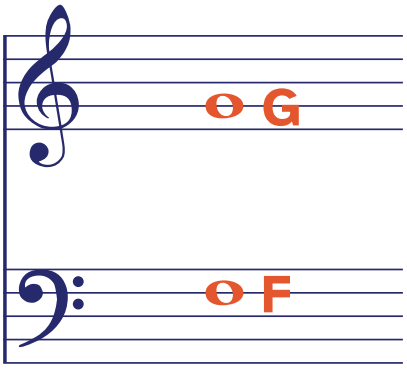
PART I · PITCH · 2

The staff and the clefs

Music is written on a **staff**: five lines and four spaces, counted from the **bottom**. Higher on the staff means higher on the keyboard. The **clef** at the start is the key to the code: it names one line, and every other note follows from it.

THE SECRET

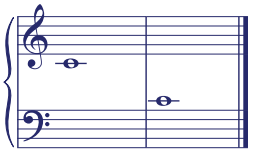
The treble clef curls round G; the bass clef's two dots hug F.



The treble clef curls round G · the bass clef's dots hug F

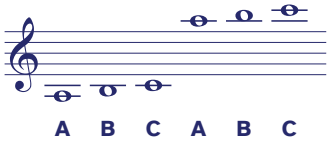
The grand staff

Piano music uses two staves joined by a brace: the **treble** staff on top, usually for the right hand, and the **bass** staff below, usually for the left. **Middle C** sits between them on a short line of its own, and can be written in either staff.

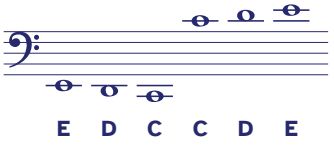


Middle C in the treble staff (left) and in the bass staff (right): the same key.

Ledger lines: the staff carried on



Treble: A, B, C below; A, B, C above.



Bass: E, D, C below; C, D, E above.

Notes too high or too low for the staff sit on short extra lines called **ledger lines**, spaced exactly like the staff lines. Read them by stepping on from the nearest staff line: line, space, line, space.

PART I · PITCH · 2 · CONTINUED

8va: play it an octave higher

Lots of ledger lines are hard to read. So a very high passage is written on the staff with **8va** and a dotted line above it: play those notes **one octave higher**, eight notes up, until the dotted line stops. **8vb** under the staff means an octave lower.

THINK OF IT A clef is a pair of glasses: the same five lines read as different notes, depending on which pair you put on.

EXAMPLE 1

One line, two notes

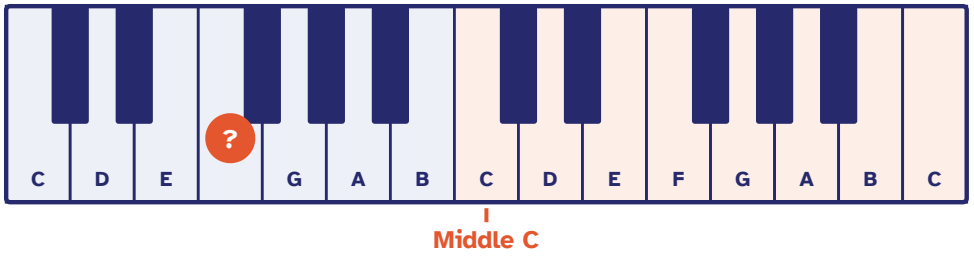


1. **Look at the red notes:** both sit on the bottom line.
2. **Now look at the clefs:** treble on the left, bass on the right.
3. **Read each from its clef note:** treble G is line 2, so step down (F in the space) to line 1; bass F is line 4, so step down line by line.

Treble bottom line: **E**. Bass bottom line: **G**. Same line, different note: always read the **clef** first.

EXAMPLE 2

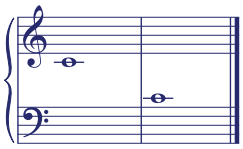
Which staff is this key written on?



1. **Find Middle C.**
2. **Is the ? to its left or right?** Left is lower: the blue half.
3. **Low keys go on the bass staff**, high keys on the treble staff.

The **?** is below Middle C, so it is written on the **bass staff**, usually for the left hand.

TRY IT · 10 SECONDS



Point, then play.

Point to the line the treble clef curls round and play that G. Point to the line between the bass clef's dots and play that F. Middle C sits between them.

PART I · PITCH · 2 · CONTINUED

EXAMPLE 3

Read a high note on ledger lines



1. **The top line** of the treble staff is F.
2. **Step up** one note at a time: G sits on top of the staff, A on the first ledger line.
3. **Keep stepping** to the red note: B above it, then the second ledger line.

Two ledger lines above the treble staff: **High C**, two octaves above Middle C.

GAME

Treble or bass?

1	2	3	4
the E above Middle C	the lowest key on the piano	Middle C	High C

Which staff would each note be written on?

Answer: 1 treble · 2 bass, with ledger lines · 3 either: it sits between the two · 4 treble, two ledger lines up.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The treble clef curls round **line 2: G**, the very note Twinkle leaps up to. Middle C hangs on its own ledger line below.

Try it. Point to the G line, then play C C G G A A G.

TRAP

- × Reading bass-clef notes as if they were treble: check the clef on every line.
- × Reading ledger lines from the wrong end: start at the staff and step outwards.

SEE ALSO
Reading the notes p. 28 · The staff and the keyboard p. 31

PRACTISE IT
Workbook Pitch 2 · Online: Library › Staff and clefs

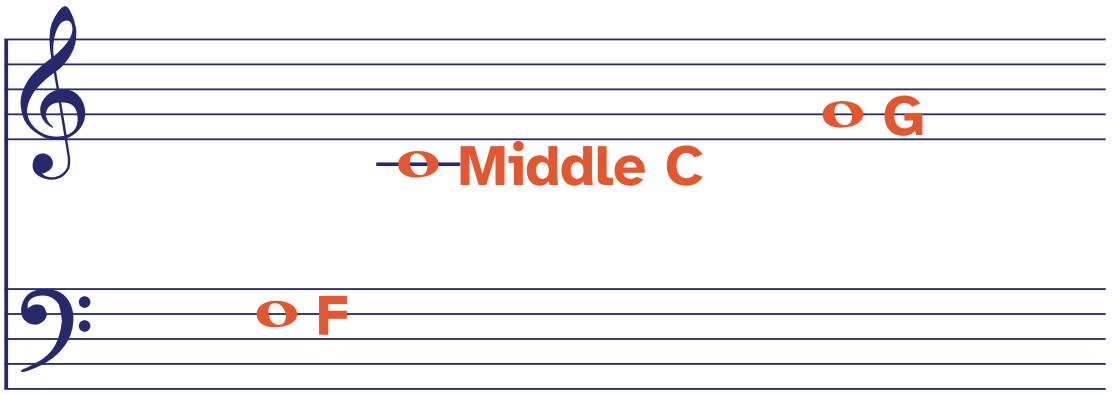
PART I · PITCH · 3

Reading the notes

Every line and space has one letter. You do not have to learn them all at once: start from a note you always know, then **step** or **skip** to the one you want.

THE SECRET

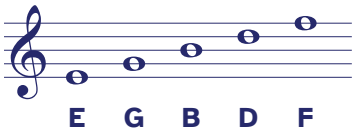
Start from a note you know (bass F, Middle C, treble G), then step or skip.



Three notes you always know: bass F, Middle C, treble G

THE RULE

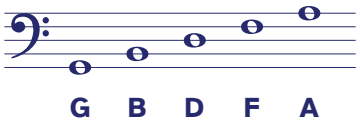
Step: to the very next line or space (one letter on). Skip: line to the next line, or space to the next space (two letters on).



Treble lines, E G B D F: *Every Good Boy Deserves Fun*



Treble spaces, F A C E: they spell *FACE*



Bass lines, G B D F A: *Great Big Dragons Fly Away*

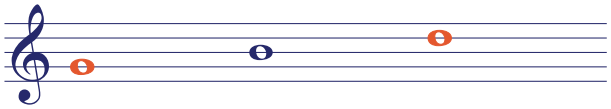


Bass spaces, A C E G: *All Cows Eat Grass*

PART I · PITCH · 3 · CONTINUED

EXAMPLE 1

Read a note you are not sure of



1. **The clef:** treble, so the clef note is G, on line 2.
2. **From G, skip up** line to line: one letter jumped each time.
3. **Count the skips** up to the other red note, on line 4: G, then B, then …

The red note is **D**: G, B, D, two skips up from the clef's G.

EXAMPLE 2

Read a bass note from the clef



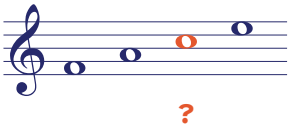
F E ?

1. **The bass clef's dots** sit either side of line 4: that line is F.
2. **Step down** to the space below.
3. **One more step down** to the red note on line 3.

F, E, **D**: bass line 3 is D, two steps down from the clef's F.

EXAMPLE 3

Find Treble C on the staff

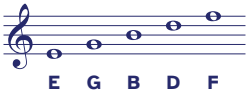


?

1. **These notes sit in the spaces.** The bottom space is F.
2. **Step up through the spaces:** F, A … they spell a word.
3. **The red note is space 3.** Which letter is it?

F A **C** E: space 3 is **C**, Treble C, the same C you found on the keyboard, one octave above Middle C.

TRY IT · 10 SECONDS



E G B D F

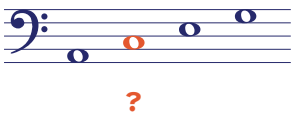
Say the treble lines up and down.

"E G B D F", then "F D B G E", out loud, as fast as you can without a slip. Then the spaces: "F A C E".

PART I · PITCH · 3 · CONTINUED

EXAMPLE 4

Find Bass C on the staff



1. **These notes sit in the bass spaces.** The bottom space is A: *All.*
2. **Say the next word:** *All Cows …*
3. **The red note is space 2.** Which letter?

A **C** E G: space 2 of the bass staff is **C**, Bass C, one octave below Middle C.

GAME

Name it from the clef note

1	2	3	4
treble line 3	treble space 1	bass line 3	bass space 3

Start from the clef note (treble G on line 2, bass F on line 4), then step or skip.

Answer: 1 B (skip up from G) · 2 F (step down from G) · 3 D (two steps down from F) · 4 E (step down from F).

YOU KNOW THIS

Mary Had a Little Lamb



E D C: **line, space, ledger line**, one step down each time. Read the shape and the tune walks down with it.

Try it. Read the notes aloud before you play them.

THINK OF IT The clef notes are lamp posts on a dark road: walk from the nearest one and you are never lost.

TRAP

- × Naming bass notes with treble names: in the bass, the bottom line is G, not E.
- × Mixing up step and skip: a step changes line to space; a skip stays on lines (or on spaces).

SEE ALSO

The staff and the clefs p. 25 · The staff and the keyboard p. 31

PRACTISE IT

Workbook Pitch 2 · Online: Library › Staff and clefs

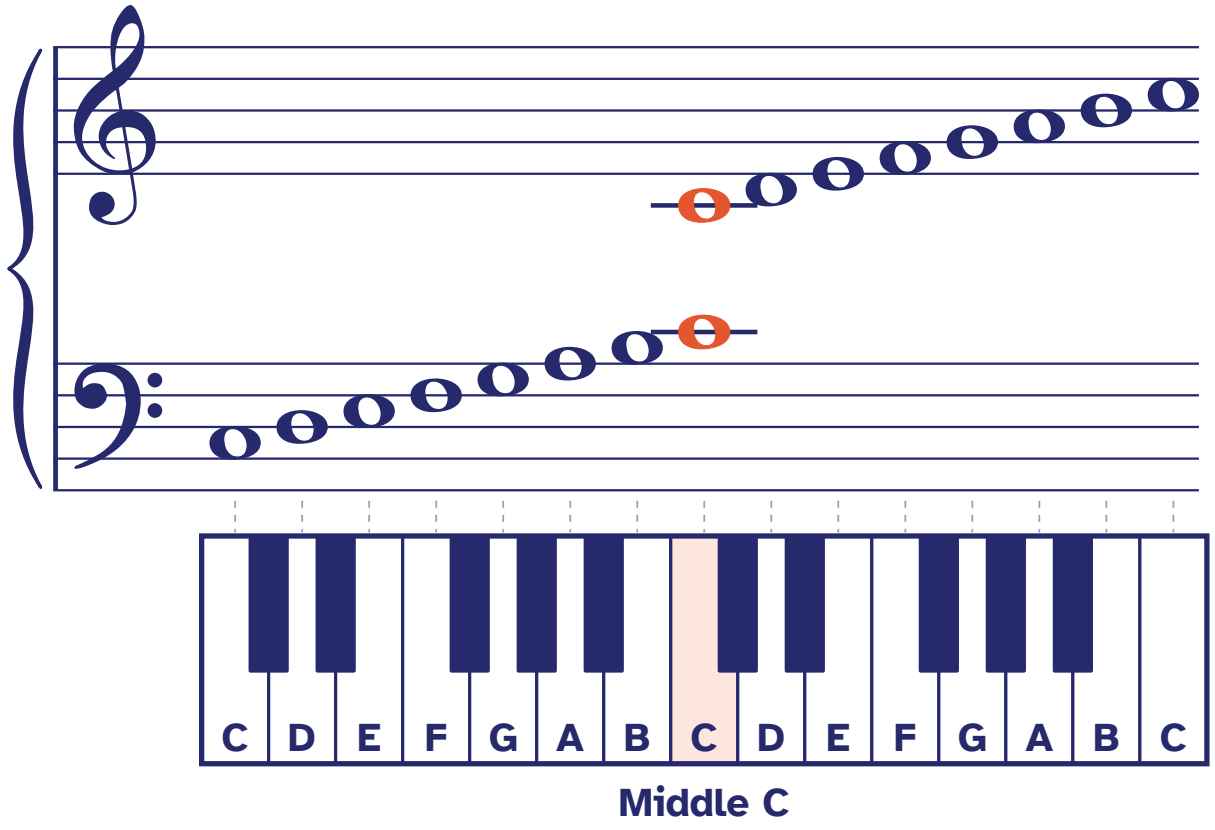
PART I · PITCH · 4

The staff and the keyboard

Each line and each space of the grand staff is one white key. Climb the staff and you climb the keyboard: here are two octaves, from Bass C to Treble C, with Middle C in red where the two staves meet.

THE SECRET

Every line and every space is one white key, in order.



Middle C can be written in either staff: on a ledger line below the treble staff, or above the bass staff. Same key; the hand that plays it chooses.

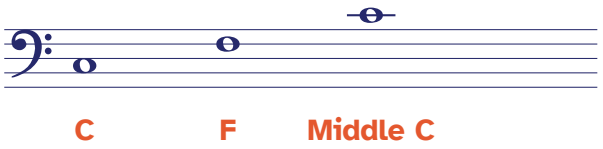
THE RULE

One step on the staff (line to space, or space to line) is the very next white key. Black keys have no line of their own: they borrow a white key's line with a sharp or flat (p. 34).

PART I · PITCH · 4 · CONTINUED

EXAMPLE 1

From the page to the key

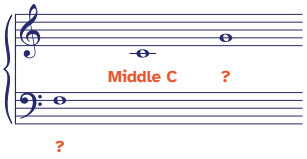


1. **Name the note:** line 4 of the bass staff, between the clef's dots, is F.
2. **Find Middle C** on the piano, then count the notes down: C is 1, B 2, A 3 ...
3. **Stop on 5** and play it with the left hand.

C B A G **F**: the bass-clef F is note 5 counting down from Middle C.

EXAMPLE 2

Mirror notes either side of Middle C

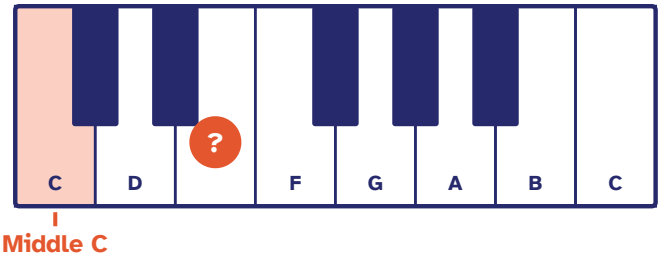


1. **Treble G:** count up from Middle C, C D E F G. Which number?
2. **Bass F:** count down from Middle C, C B A G F. Which number?
3. **Compare the two.**

Both are **note 5**: treble G and bass F are mirror images, the same distance either side of Middle C.

EXAMPLE 3

From the key to the page



1. **Find the ? key:** count up from Middle C, C 1, D 2 ...
2. **Do the same on the staff:** Middle C on its ledger line, D in the space under the staff ...
3. **Where do you land?**

The **?** is **E**, note 3, and it sits on the **bottom line** of the treble staff.

PART I · PITCH · 4 · CONTINUED

GAME

Count the notes

1	2	3
Middle C up to A	Middle C down to E	G up to the next G

Count the notes, both ends included. The staff climbs the same way: one line or space per note.

Answer: 1 six (C D E F G A) · 2 six (C B A G F E) · 3 eight: an octave.

TRY IT · 10 SECONDS

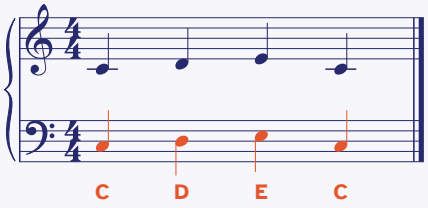


Play Middle C, then walk up.

Say each white key as you play it, C D E F G A B C, and picture each one climbing the staff: line, space, line, space.

YOU KNOW THIS

Frère Jacques, in both hands



The left hand plays the **same letters**, C D E C, one octave lower, written in the bass clef.

Try it. Play it right hand, then left hand, then both together.

THINK OF IT The staff is a ladder laid beside the keyboard: each rung, a line or a space, is the next white key.

TRAP

- × Thinking the black keys have their own lines: they borrow a white key's line or space with a sharp or flat.
- × Counting from 0: the note you start on is note 1.

SEE ALSO

The keyboard p. 22 · Reading the notes p. 28

PRACTISE IT

Workbook Pitch 1–2 · Online: Library › Staff and clefs

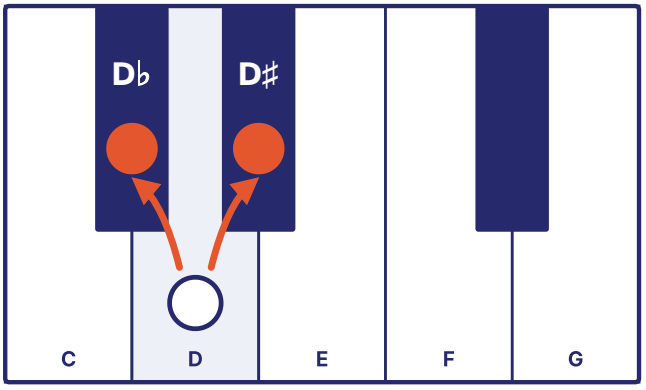
PART I · PITCH · 5

Sharps, flats and naturals

A sign in front of a note moves it to the very next key. **Sharp**: one key to the **right**. **Flat**: one key to the **left**. Usually that next key is black, which is how the black keys get their names.

THE SECRET

Sharp: the very next key to the right. Flat: the very next key to the left.

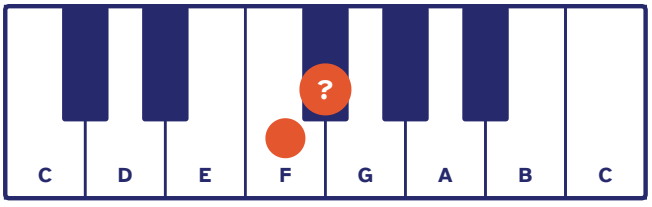


♯ one key right · ♭ one key left

SIGN	NAME	WHAT IT DOES	FIRST MET
♯	sharp	one key to the right (a semitone higher)	Level 0
♭	flat	one key to the left (a semitone lower)	Level 0
♮	natural	cancels a sharp or flat: back to the white key	Level ½

EXAMPLE 1

Find F♯



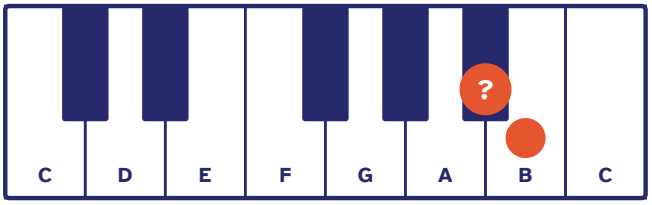
1. **Find F** (the red dot): just left of the three black keys.
2. **Sharp means right:** move to the very next key.
3. **Is it black or white?** Put your finger on it.

The **?** is **F♯**: the first black key of the group of three.

PART I · PITCH · 5 · CONTINUED

EXAMPLE 2

Find B♭



1. **Find B** (the red dot): just right of the three black keys.
2. **Flat means left:** move to the very next key.
3. **Which black key is it?**

The **?** is **B♭**: the top black key of the group of three.

EXAMPLE 3

How long does the sharp last?

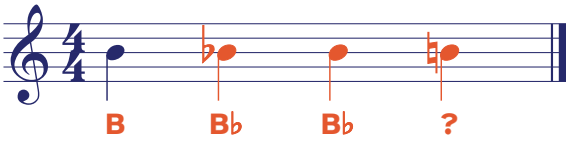


1. **The first note** has a sharp: F♯.
2. **Later in the same bar** there is no sign. Is it still sharp?
3. **After the bar line**: is it still sharp?

Still **F♯** until the bar line; after it, plain **F** again. A sign lasts **to the end of its bar**.

EXAMPLE 4

A natural cancels the flat



1. **B**: the white key.
2. **B♭**: one key left, and it stays flat for the rest of the bar.
3. **The last note** has a natural sign: what does it do?

The natural **cancels** the flat: back to the white key B, until the bar line.

PART I · PITCH · 5 · CONTINUED

EXAMPLE 5

One key, two names



1 **The first bar:** F♯, then G♭. Find both on the keyboard.

2 **Are they different keys?**

3 **The last bar:** E♯. Which key is one right of E?

F♯ and G♭ are the **same black key**; E♯ is the white key **F**. Same sound, different names: **enharmonic**.

GAME

Which key is it?

1	2	3	4
C♯	E♭	B♯	F♭

Sharp: one key to the right. Flat: one key to the left. Which key do you press?

Answer: 1 the black key right of C · 2 the black key left of E · 3 the white key C · 4 the white key E.

YOU KNOW THIS

Für Elise (Beethoven)



E, D♯, E, D♯: the very next key down and back. The sharp is written again in the new bar, because the bar line wiped it out.

Try it. Play E and D♯ back and forth: the black key just left of E.

THINK OF IT A sharp or flat is a sticky note on one line of the staff: it stays stuck until the bar line, then it falls off.

TRAP

× Carrying a sharp or flat past the bar line: after it, the note is plain again.

SEE ALSO
Keys and key signatures p. 75

PRACTISE IT
Workbook Pitch 3 · Online: Library › Signs

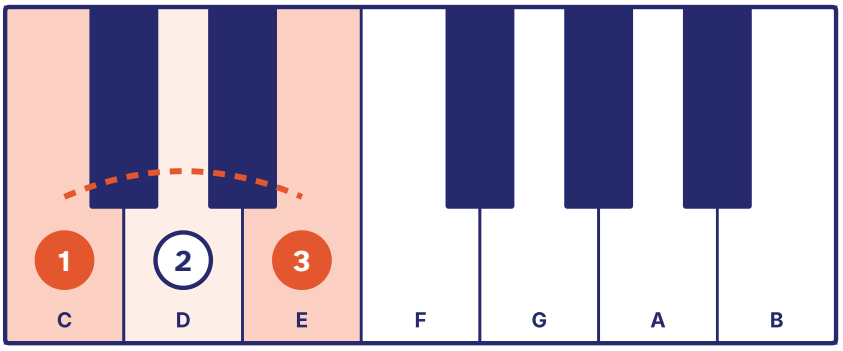
PART I · PITCH · 6

Counting intervals

How far apart are two notes? That distance is an **interval**. To give it a number, **count the notes**: the first note is 1, and you count every letter up to the second, both ends included.

THE SECRET

Count the notes, both ends included: C D E is 3 notes, a 3rd.

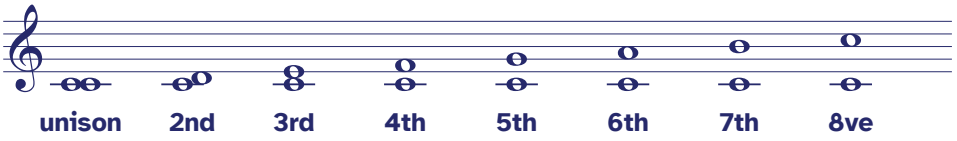


C to E: count the notes, 1 2 3: a 3rd

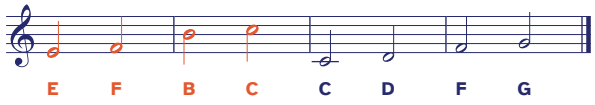
THE RULE

Count the notes for the number: start at 1 on the first note. C to E is C D E: 3 notes, a 3rd. Sharps and flats never change the number.

Interval numbers from C



The smallest steps: semitones and tones



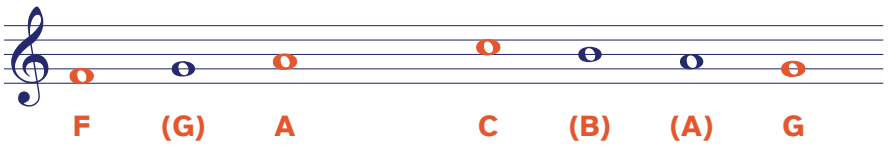
E–F and B–C: no black key between, a semitone. C–D and F–G: a black key between, a tone.

From any key to the **very next key**, black or white, is a **semitone**: the smallest step on the piano. Two semitones make a **tone**. On the white keys most steps are tones, but **E–F** and **B–C** have no black key between them: they are semitones.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 1

How far apart are these notes?

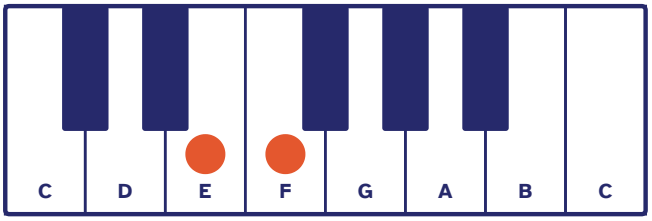


1. **Call the first note 1:** here F is 1.
2. **Count every letter** up to the second note.
3. **Going down works the same:** start at 1 on the top note and count down.

F G A is 3 notes: a **3rd**. C B A G is 4 notes: a **4th**.

EXAMPLE 2

Tone or semitone?

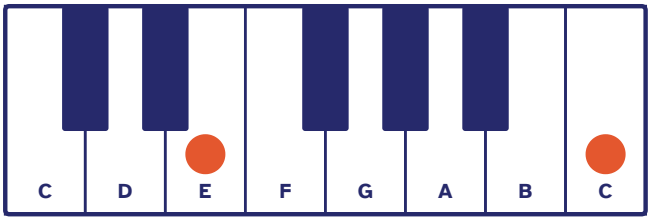


1. **Look between the red keys:** is there a black key?
2. **No black key** means they are next-door keys.
3. **Now try C and D**: is there a black key between?

E–F is a **semitone** (next door). C–D has C♯ between: a **tone**.

EXAMPLE 3

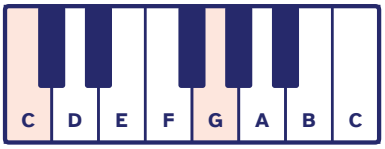
Name an interval on the keyboard



1. **Find both red keys:** E, and the C above it.
2. **Count the notes** from E: E is 1 …
3. **Stop on the C.** What number is it?

E F G A B C is 6 notes: a **6th**.

TRY IT · 10 SECONDS



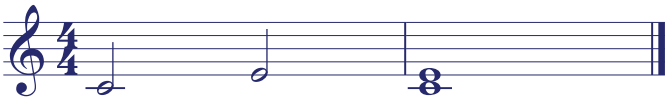
Thumb on C, finger 5 on G.

Your hand is a ruler: thumb to little finger on white keys is a 5th. Play it, then thumb to finger 3: a 3rd.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 4

Melodic or harmonic?



1 **The first bar:** C, then E, one after the other.

2 **The second bar:** C and E at the same time.

3 **Count the notes** in each.

Both are a **3rd**: **melodic** when played one after the other, **harmonic** when played together.

GAME

Count the notes

1	2	3	4
C up to G	D up to F	E up to the next E	G up to A

Count every note, both ends included. What is each interval called?

Answer: 1 a 5th (C D E F G) · 2 a 3rd · 3 an octave (8) · 4 a 2nd.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The leap from the first “twinkle” to the second is C up to G: C D E F G, **five notes, a 5th**.

Try it. Sing the first two notes of any song, then count the notes between them.

THINK OF IT Monday to Wednesday is three days, not two: you count the day you start on. Intervals count the same way.

TRAP

- × Counting only the gaps, not the notes: C to E is 3 notes, not 2 gaps.
- × Letting a sharp or flat change the number: C to G♯ is still a 5th.

SEE ALSO

Major, minor and perfect p. 40 · Major scales p. 69

PRACTISE IT

Workbook Pitch 4 · Online: Library › Intervals

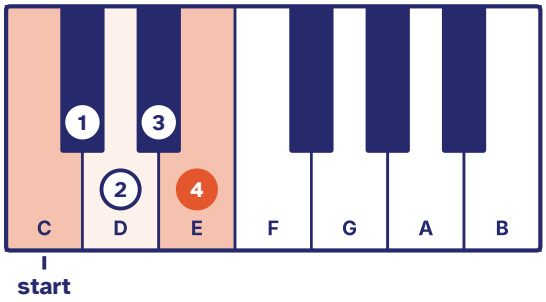
PART I · PITCH · 7

Major, minor and perfect

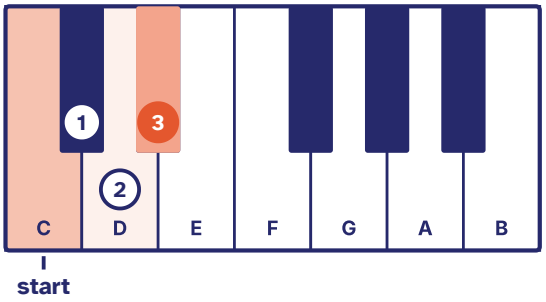
C to E and C to E♭ are both 3rds: three notes each. But they sound different, bright and dark. To tell them apart, **count the hops**: every move to the very next key, black or white, is one hop.

THE SECRET

Count the hops: 4 make a major 3rd, 3 a minor 3rd.



C to **E**: 4 hops. A **major** 3rd: bright.



C to **E♭**: 3 hops, one fewer. A **minor** 3rd: dark.

One hop less: a major 3rd becomes minor

THE RULE

Count the notes for the number (a 3rd). Count the hops for the size (major or minor). Do not count the key you start on: the first move is hop 1.

NUMBER	SMALLER	MAIN SIZE
3rd	3 (minor)	4 (major)
4th		5 (perfect)
5th		7 (perfect)
6th	8 (minor)	9 (major)
octave		12 (perfect)

3rds and 6ths come in two sizes: **major**, and **minor**, one hop smaller.

4ths, 5ths and octaves have one main size, called **perfect**. They are never major or minor.

EXAMPLE 1

Major or minor 3rd?



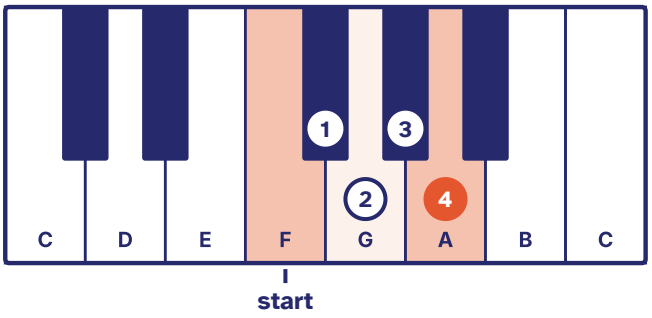
1. **Count the notes:** C D E, three letters, so it is a 3rd.
2. **Count the hops** from C to the red black key, E♭: C♯ 1, D 2 …
3. **Major has 4 hops.** How many here?

C to E♭ is **3 hops**: a **minor 3rd**.

PART I · PITCH · 7 · CONTINUED

EXAMPLE 2

Name F up to A

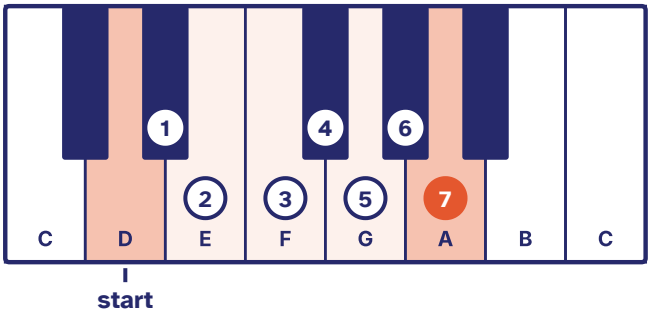


1. **Count the notes:** F G A is 3, so it is a 3rd.
2. **Count the hops:** every key you land on, black and white, is numbered.
3. **4 hops or 3?**

F to A is 4 hops: a **major 3rd**, the same size as C to E.

EXAMPLE 3

Name D up to A

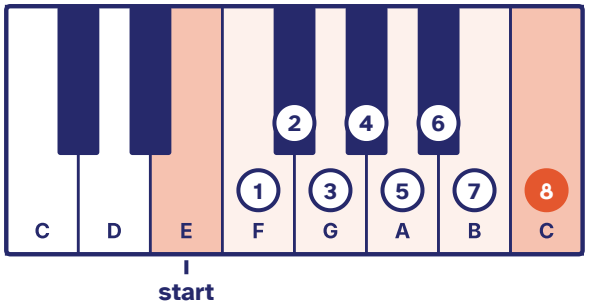


1. **The number:** D E F G A, five notes, a 5th.
2. **The size:** count the hops: the numbered keys.
3. **A 5th of 7 hops** is its main size. What is that called?

D up to A is a **perfect 5th**: 7 hops.

EXAMPLE 4

Name E up to C



1. **The number:** E F G A B C, six notes, a 6th.
2. **The size:** count the hops.
3. **A major 6th** has 9 hops. Is this one more or one fewer?

E up to C is 8 hops, one fewer than major: a **minor 6th**.

PART I · PITCH · 7 · CONTINUED

GAME

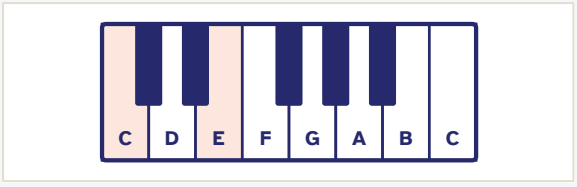
Perfect, major or minor?

1	2	3	4
C–G	D–F	C–E	E–A

Count the notes for the number, then the hops for the size.

Answer: 1 perfect 5th (7 hops) · 2 minor 3rd (3) · 3 major 3rd (4) · 4 perfect 4th (5).

TRY IT · 10 SECONDS

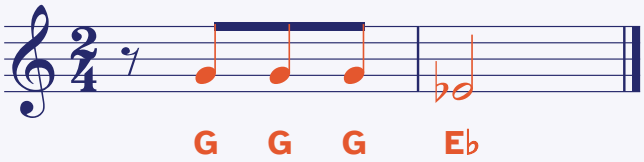


Play C–E, then C–E♭.

Bright, then dark. One hop makes the difference: the major 3rd has four, the minor 3rd three.

YOU KNOW THIS

Beethoven, Symphony No. 5



Da-da-da-DUM: G down to **E♭** is three notes, a 3rd, and 4 hops (F♯, F, E, E♭): a **major 3rd**.

Try it. Count the hops from G down to E♭ on the keyboard.

THINK OF IT A major and a minor 3rd are a smile and a frown: the same face, one small change at the corners.

TRAP

- × Counting white keys only: count every hop, black and white.
- × Counting the start key as hop 1: the first move is hop 1.
- × Calling a 4th or 5th “major”: they are perfect.

SEE ALSO

Counting intervals p. 37 · Chords and triads p. 86

PRACTISE IT

Workbook Pitch 4 · Online: Library › Intervals

PART I · PITCH · TIPS

When a note reads wrong

Most wrong notes come from a handful of slips. Name the slip first, then fix only that note: playing the whole line again rarely finds it.

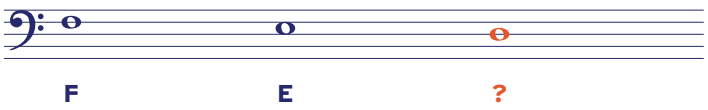
TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
The left hand reads like the right	**the clef**: bass lines are G B D F A, not E G B D F	p. 25
You land an octave too high or too low	**the Cs**: find Middle C first, then walk out to Bass C or Treble C	p. 22
Notes above or below the staff slow you down	**the ledger lines**: step on from the nearest staff line	p. 25
A sharp or flat vanishes, or stays too long	**the bar line**: a sign lasts only to the end of its bar	p. 35
You play the black key on the wrong side	**the direction**: sharp goes right, flat goes left	p. 34
Two notes sound too close or too far apart	**the interval**: count the notes, then the hops	p. 40

EXAMPLE

Spot the slip



Sam played B. Wrong!

1. **Check the clef first:** this is the bass clef, not the treble.
2. **Start from the clef note:** the dots hug line 4, F.
3. **Step down** to the red note on line 3: F, E ...

The note is **D**. Sam read the bass staff with treble names (treble line 3 is B): the most common slip of all.

HOW TO

Three checks before you read a new line

1. **The clefs:** which staff is treble, which is bass?
2. **The clef notes:** find treble G, bass F and Middle C, and read the other notes from them.
3. **The signs:** mark each sharp, flat or natural, and see where its bar ends.

THINK OF IT Like reading a map: find the big landmarks first, then the streets around them. A note far from every clef note is the one most often misread.